segments over the cartesian axes don’t intersect both on the x and y axis.

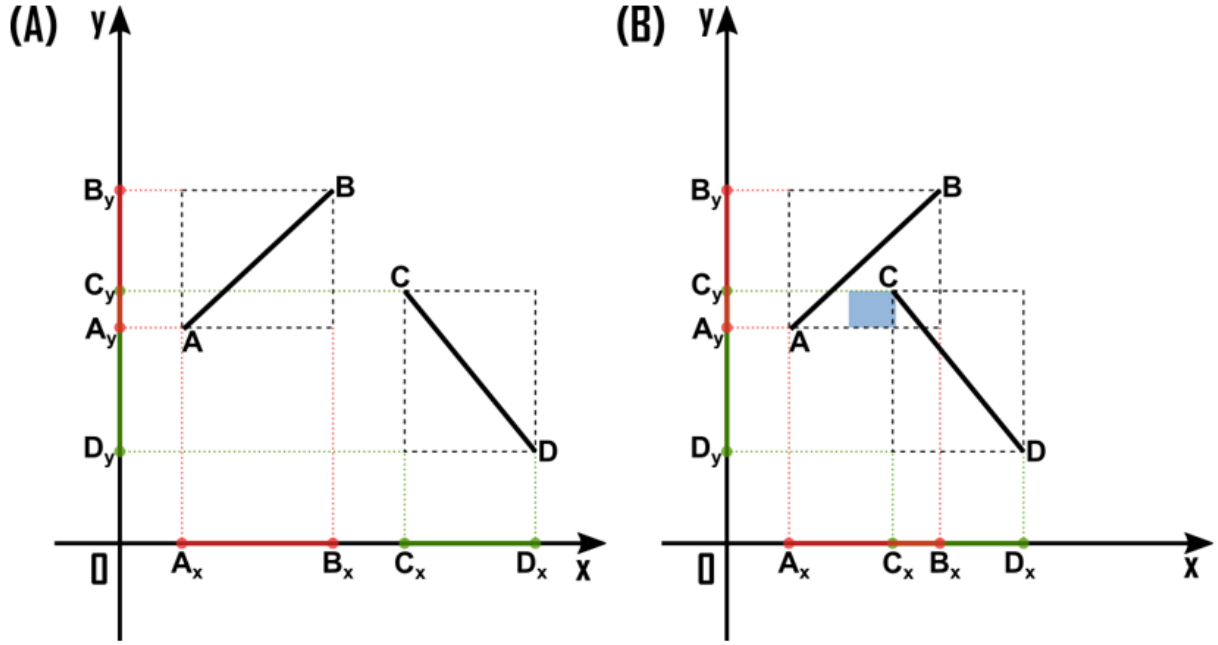


Figure 15.16 Examples of non-intersecting segments: the bounding-box check can only rule out an intersection, but cannot confirm it.

This condition, having the projected segments not intersecting, is however sufficient, but not necessary for establishing that two segments do not cross: 15.16.B shows how there can be segments whose boxes intersect, but that doesn’t cross.